The 5 AI Terms Everyone Should Know in 2026

Introduction

Artificial Intelligence is evolving faster than ever. Every week, new AI models, tools, and technologies are launched. But behind all the hype, there are a few important concepts that power modern AI systems.

If you understand these 5 terms, you already know more than most beginners entering AI today.

In this document, we will learn:

1. SFT (Supervised Fine-Tuning)
2. RLVR (Reinforcement Learning with Verifiable Rewards)
3. Distillation
4. GGUF
5. Off-Policy Learning

This guide explains each topic in simple English with examples, workflows, use cases, and useful learning resources.

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1. SFT — Supervised Fine-Tuning

What is SFT?

SFT stands for Supervised Fine-Tuning.

It is the process of training a pre-trained AI model using labeled examples created by humans.

A large language model already knows general knowledge from internet-scale data. SFT helps the model behave in a more useful, safe, and task-specific way.

Examples:

- Chatbots
- Coding assistants
- AI customer support
- AI writing tools

Simple Example

Input:
"Write an email for leave approval."

Expected Output:
"Dear Manager, I would like to request leave…"

The AI learns from thousands of these examples.

SFT Workflow

Pretrained Model
↓
Human-Created Dataset
↓
Fine-Tuning Process
↓
Specialized AI Assistant

Why SFT is Important

Benefits:

- Makes AI more accurate
- Improves response quality
- Adds domain-specific knowledge
- Helps AI follow instructions better

Real-World Use Cases

- AI coding assistants
- Medical chatbots
- AI teachers
- Legal document assistants
- Customer support automation

OpenAI SFT Guide

Official Documentation:
https://developers.openai.com/api/docs/guides/supervised-fine-tuning

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2. RLVR — Reinforcement Learning with Verifiable Rewards

What is RLVR?

RLVR stands for Reinforcement Learning with Verifiable Rewards.

It is a modern AI training method where the model gets rewards only when outputs can be verified as correct.

Instead of humans manually rating responses, the system automatically checks correctness.

Example

Question:
"What is 25 × 12?"

AI Output:
300

The answer can be verified automatically.

Correct Answer → Reward
Wrong Answer → No Reward

Why RLVR Matters

RLVR is useful for:

- Mathematics
- Coding
- Logical reasoning
- Scientific problems

Because these tasks have verifiable answers.

RLVR vs RLHF

RLHF:
Humans rate responses manually.

RLVR:
The system verifies correctness automatically.

RLVR is faster and cheaper for scalable AI training.

RLVR Workflow

Question
↓
AI Generates Answer
↓
Automatic Verification
↓
Reward or Penalty
↓
Model Improves

Applications

- AI coding models
- Math-solving agents
- Autonomous AI systems
- Research agents

Learning Resource

Awesome RLVR Repository:
https://github.com/opendilab/awesome-RLVR

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3. Distillation

What is Distillation?

Distillation is a technique where a large AI model teaches a smaller model.

The smaller model becomes faster and cheaper while keeping most of the intelligence.

Teacher Model → Student Model

Why Distillation is Important

Large models are expensive.

Distillation helps:

- Reduce memory usage
- Improve speed
- Run AI on laptops and phones
- Lower GPU costs

Simple Analogy

Imagine a professor teaching important concepts to a student in simplified form.

The student learns faster while keeping the essential knowledge.

Distillation Workflow

Large Model
↓
Generate Knowledge
↓
Train Smaller Model
↓
Efficient Lightweight Model

Benefits

- Faster inference
- Lower hardware requirements
- Better deployment efficiency
- Reduced energy consumption

Real-World Examples

- Mobile AI assistants
- Edge AI devices
- Lightweight chatbots
- Local AI applications

GitHub Distillation Projects

Explore Repositories:
https://github.com/topics/distillation

Additional Learning Material

MIT Distillation Handout:

https://ocw.mit.edu/courses/5-301-chemistry-laboratory-techniques-january-iap-2012/623aab3011076b0f06d687b73bf6446c_MIT5_301IAP12_Dist_Handout.pdf

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4. GGUF

What is GGUF?

GGUF stands for GPT-Generated Unified Format.

It is a file format used for running Large Language Models locally.

GGUF is especially popular in local AI communities.

Why GGUF Matters

GGUF allows:

- Faster loading
- Efficient memory usage
- CPU-based inference
- Quantized AI models

This makes local AI more practical.

Tools That Use GGUF

- llama.cpp
- Ollama
- LM Studio
- KoboldCpp

What is Quantization?

Quantization reduces model size.

Example:
A 70GB model can become:

- 40GB
- 20GB
- Even smaller

This helps run AI on normal computers.

GGUF Workflow

Original AI Model
↓
Quantization
↓
Convert to GGUF

↓
Run Locally

Advantages of GGUF

- Run AI offline
- Lower RAM usage
- Faster inference
- Easier deployment

Learning Resource

GGUF Learning Guide:
https://github.com/nameissainath/gguf-learning-guide

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5. Off-Policy Learning

What is Off-Policy Learning?

Off-Policy Learning is a reinforcement learning technique where an AI learns from previously collected experiences.

The model does not need fresh data every time.

Simple Example

Imagine learning driving skills by watching recorded driving videos instead of driving yourself.

That is similar to off-policy learning.

Why It Is Useful

Benefits:

- Reuses old data
- Saves computation cost
- Improves training efficiency
- Enables large-scale learning

Off-Policy vs On-Policy

On-Policy:
Learns only from current actions.

Off-Policy:
Learns from old and external experiences.

Popular Algorithms

- DQN
- SAC
- TD3
- Q-Learning

These algorithms are widely used in modern reinforcement learning.

Applications

- Robotics
- Self-driving cars
- Game AI
- AI agents
- Recommendation systems

Learning Resource

Comprehensive Guide:
https://www.numberanalytics.com/blog/off-policy-learning-comprehensive-guide

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Final Thoughts

These 5 concepts represent the foundation of modern AI systems.

SFT teaches AI using human examples.

RLVR improves AI using verifiable rewards.

Distillation makes models smaller and faster.

GGUF enables local AI execution.

Off-Policy Learning helps AI learn efficiently from past experiences.

Understanding these topics gives you a strong advantage in AI engineering, model training, and future AI development.

Recommended Next Steps

After learning these concepts, explore:

- LLM Fine-Tuning
- Quantization
- AI Agents
- Reinforcement Learning
- Local AI Deployment
- AI Infrastructure

All Resources

1. OpenAI SFT Guide
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2. Awesome RLVR Repository
   https://github.com/opendilab/awesome-RLVR
3. MIT Distillation Handout

   https://ocw.mit.edu/courses/5-301-chemistry-laboratory-techniques-january-iap-2012/623aab3011076b0f06d687b73bf6446c_MIT5_301IAP12_Dist_Handout.pdf
4. Distillation GitHub Projects
   https://github.com/topics/distillation

5. GGUF Learning Guide
   https://github.com/nameissainath/gguf-learning-guide
6. Off-Policy Learning Guide
   https://www.numberanalytics.com/blog/off-policy-learning-comprehensive-guide